

VAIAKU, Tuvalu

WMO: 916430

Lat: 8.522S

Lon: 179.198E

Elev: 7

StdP: 14.69

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 80705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	76.6	77.1	71.2	115.2	82.1	71.7	117.2	82.2	27.4	82.2	25.3	82.6	8.4	20	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	7.8	90.8	81.4	89.9	80.9	89.7	80.7	82.8	88.3	82.4	88.0	81.9	87.5	8.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.3	87.0	80.7	159.7	86.8	80.3	157.0	86.4	46.9	89.4	46.4	88.2	45.7	87.7	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.3	21.2	18.8	73.9	92.4	1.6	0.6	72.8	92.8	71.8	93.2	70.9	93.5	69.7	94.0		
(5)			71.6	85.3	0.9	1.4	71.0	86.3	70.4	87.1	69.9	87.9	69.2	88.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.0	83.7	83.5	83.7	84.3	84.3	84.2	83.9	83.6	84.0	84.1	84.2	84.1
	DBStd	1.82	2.09	1.85	1.91	1.83	1.75	1.66	1.67	1.77	1.62	1.79	1.80	1.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12401	1045	939	1043	1029	1063	1026	1052	1043	1021	1057	1026	1057
	CDD65	6926	580	519	578	579	598	576	587	578	571	592	576	592
	CDH74	84660	7115	6230	7000	7176	7384	7086	7062	6886	6975	7306	7103	7337
	CDH80	33171	2782	2310	2669	2937	2984	2824	2673	2518	2714	2935	2865	2960
Wind	WSAvg	8.4	8.8	9.4	7.5	7.4	7.2	9.4	10.1	9.4	9.3	7.3	7.2	7.9
Precipitation	PrecAvg	136.5	16.1	12.4	13.9	10.6	9.4	8.5	9.7	9.9	9.0	9.9	11.0	16.3
	PrecMax	190.3	35.1	29.5	38.5	24.4	24.2	16.4	22.2	26.4	26.8	19.0	26.0	33.0
	PrecMin	87.6	6.6	3.7	5.8	2.5	1.8	3.7	2.8	3.7	2.6	2.4	2.0	6.0
	PrecStd	24.3	6.2	6.1	6.0	5.6	4.6	3.7	4.4	4.8	5.3	3.9	5.4	7.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.2	90.0	90.4	91.0	90.5	90.3	90.0	89.9	90.0	90.5	91.4	91.5
		MCWB	81.6	80.8	81.0	81.7	81.4	81.4	81.1	80.9	80.9	81.1	81.5	81.9
	2%	DB	89.8	89.4	89.7	89.8	89.7	89.6	89.4	89.2	89.6	89.7	89.9	89.9
		MCWB	80.6	80.5	80.5	81.0	81.0	81.1	80.7	80.6	80.6	80.5	80.6	80.9
	5%	DB	89.1	88.1	88.6	89.2	88.9	88.4	88.1	88.0	88.2	89.1	89.4	89.2
		MCWB	80.3	80.0	80.3	80.8	80.8	80.7	80.4	80.3	80.3	80.2	80.3	80.5
	10%	DB	87.9	87.6	87.8	88.0	87.9	87.7	87.5	87.4	87.7	88.0	88.1	87.9
		MCWB	79.9	79.8	80.1	80.5	80.6	80.5	80.1	80.1	80.1	79.9	79.8	80.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.0	82.8	82.6	82.8	82.9	82.8	82.7	82.6	83.0	82.8	83.2	83.2
		MCDB	88.1	87.8	88.4	88.7	88.0	88.1	88.0	87.6	87.8	88.4	89.0	89.1
	2%	WB	81.9	81.7	81.7	82.0	82.1	82.0	81.7	81.6	81.8	81.8	82.0	81.9
		MCDB	87.3	86.9	87.6	88.0	87.6	87.3	87.1	86.8	87.1	87.8	88.2	87.8
	5%	WB	81.1	80.9	81.0	81.4	81.5	81.4	80.9	80.9	81.0	80.9	81.1	81.2
		MCDB	86.7	86.2	86.8	87.3	86.9	86.6	86.3	86.4	86.5	87.1	87.1	87.1
	10%	WB	80.5	80.2	80.4	80.8	80.9	80.8	80.4	80.3	80.4	80.2	80.4	80.6
		MCDB	86.1	85.5	86.1	86.6	86.3	86.0	85.6	85.7	86.0	86.3	86.2	86.3
Mean Daily Temperature Range	5% DB	MDBR	7.7	7.7	7.9	7.8	7.7	7.4	7.3	7.6	7.5	8.1	8.1	8.1
		MCDBR	8.6	8.5	8.6	8.4	8.3	8.1	8.1	8.3	8.2	8.7	9.0	9.2
		MCWBR	4.8	4.7	4.7	4.4	4.5	4.6	4.4	4.7	4.7	5.0	4.9	4.8
	5% WB	MCDBR	8.0	8.0	8.2	8.0	7.8	7.6	7.5	7.8	7.8	8.3	8.6	8.5
		MCWBR	5.0	4.8	4.8	4.6	4.6	4.7	4.5	4.8	4.9	5.0	5.0	4.9
Clear-Sky Solar Irradiance	taub	0.412	0.413	0.394	0.390	0.396	0.412	0.414	0.418	0.417	0.405	0.406	0.405	
	taud	2.559	2.548	2.624	2.639	2.607	2.538	2.527	2.514	2.505	2.554	2.560	2.576	
	Ebn at Noon	295	293	294	287	276	266	268	274	283	292	294	296	
	Edh at Noon	34	35	31	30	29	30	31	33	35	34	34	34	
All-Sky Solar Radiation	RadAvg	1698	1687	1794	1732	1552	1464	1515	1677	1881	1907	1800	1743	
	RadStd	190	205	157	167	107	102	121	101	122	153	139	128	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page